Mohsen Dirbaz

Chicago, Illinois, US
Interview Availability: 2 weeks
Start Availability: January 2026
Education: Ph.D. Chemical Engineering / M.S. Natural Gas Engineering / B.Sc. Petroleum Engineering
Latest Position: Postdoctoral Fellow, University of Missouri-Columbia
Work Authorization: Green Card

Candidate Overview / Highlighted Skills

- A well-rounded Chemical Engineer with 5+ years in-depth knowledge of process design, modeling, and simulation in oil & gas industries, ASPEN Plus, HYSYS, PFD's, P&ID's, and heat and material balances.
- Strong foundation in process engineering with hands-on experience in core unit operations (e.g., distillation, heat exchange, reactors) and supporting equipment such as pumps, compressors, and separation systems.
- A solid combination of academic excellence, technical proficiency, and practical project experience, complemented by a proactive, self-motivated, and detail-oriented approach to individual and collaborative endeavors.
- Served as a Process Engineer at L&T Technology Services, where responsibilities included review of PFDs and PIDs, fluid flow calculations, and coordination with equipment vendors and fabrication partners.
- Technical capabilities span advanced simulation and programming tools such as Aspen Plus, HYSYS, MATLAB, Mathematica, Python, JavaScript, NumPy, and Jupyter notebooks.
- Recognized with multiple honors and awards, including Teaching Assistant of the Year at Illinois Institute of Technology and competitive graduate assistantships.
- Active participation in professional organizations such as AIChE and SPE, as well as leadership roles in academic associations.

Education

Illinois Institute of Technology (IIT), Chicago, IL, USA
Ph.D. in Chemical Engineering - GPA: 3.6 (2012-2020)

- Dissertation: "A Neural Network Based Model for Biomass Gasification in Fluidized bed"
- Researched gasification and emission control technologies relevant to IGCC systems, including biomass and coal gasification in fixed, entrained, and fluidized bed reactors.
- Investigated intermediate purification and treatment steps—such as sulfur compound removal, trace metal abatement, COS hydrolysis, and carbon capture and other enabling technologies that enhance technical and economic viability of power generation and chemicals production.
- Conducted research on the environmental impacts of hydraulic fracturing and evaluated emerging frac water remediation strategies—including physico-chemical and biological treatment pathways, process models (e.g. water chemistry models for calculation of PH and precipitation reactions, mass transfer models for oxygen modeling and other gas/liquid interactions) with a focus on sustainable water management practices.

Texas A&M University-Kingsville (TAMUK), TX, USA
M.S. Natural Gas Engineering - GPA: 3.9 (2010-2012)

- Dissertation: "Evaluation of Formation Damage in Horizontally Fractured Wells in Low Permeability Reservoirs"
- Performed advanced process simulation and optimization of industrial gas processing systems using HYSYS software, including acid gas sweetening, dehydration, and fractionation units.
- Designed and optimized compressor station configurations for a 1,800-mile natural gas transmission pipeline to ensure efficient flow and minimize pressure losses.
- Performed preliminary economic analysis and material selection to determine optimal pipe diameter, grade, and specification based on maximum pressure and performance requirements.
- Conducted specialized research in offshore petroleum engineering, focusing on subsea well completion technologies, cementitious material rheology, and enhanced oil recovery screening methodologies.

Sharif University of Technology (SUT), Tehran, Iran
B.Sc. in Petroleum Engineering (2005-2010)

- Gained hands-on experience in PVT lab, drilling lab, reservoir rock, and fluid mechanics lab.
- Completed a comparative analysis for constant composition expansion and constant volume depletion.

Experience

University of Missouri-Columbia (2023–2025)
Postdoctoral Fellow

- Developed TEA Space, a platform that facilitates comprehensive techno-economic assessments by integrating dynamic modeling and simulation capabilities in a highly customizable interface.
- Conducted in-depth analysis on feasibility of manufacturing Nitrogen trifluoride, a critical material used by semiconductor foundries for wafer etching and chamber cleaning.
- Presented at AICHE: "Optimally Dispersed Network of Biomass Gasification to Produce Green H_2"
- Authored several technical reports:
  - **NH_3 Report:**
    * Gray Haber-Bosch Process. Advanced Blue Processes.
    * Carbon Capture Technologies. Hydrogen Transportation and Storage Options.
    * Green Ammonia via Electrochemical Synthesis. Lithium Mediated Ammonia Production.
  - **H_2 Report:**
    * Hydrogen Production Routes. Electrolysis. Fossil Resources. Biomass.
    * Hydrogen Powered Fleet. Expansion Strategies and Market Penetration.
    * Biomass Carbon Neutrality. Circular Economy. Life Cycle Assessment.
- Tinkered An Energy-Policy Roadmap, Introducing A Forward-Thinking Renewable Energy Blueprint Built on 6 Pillars of:
  1. Climate liability (adaptation vs mitigation)
  2. Electrifying rural economies (tribal, local, state and territorial models)
  3. Sustainable forestry (management, certification, carbon accounting)
  4. Biomass utilization (value chain pathway optimization)
  5. Hydrogen demand (incentive mechanisms)

Didas International Group/Iran (2020–2023)
Energy Initiatives Consultancy

- Reviewed Regulatory Compliance Status of Flare Treatment Technologies.
- Mapped Silica Processing End Use Applications of Reflective Coatings for Solar Thermal Systems.
- Analyzed Impact of Euro Adoption on the GDP Dynamics of Middle Eastern Countries.
- Investigated Systemic Disparity of Training-Deployment in Engineering Education in US.

Larsen and Toubro Technology Services Limited (2019)
Process Engineer

- Acted as key liaison between offshore engineers and clients, guaranteeing project alignment.
- Reviewed PFDs (Process Flow Diagrams) and P&IDs (Process and Instrumentation Diagrams).
- Delivered comprehensive progress reports at various milestones, fostering transparency and client trust.
- Collaborated with suppliers to finalize equipment designs, optimizing project outcomes.

Illinois Institute of Technology (2015–2017)
Teaching Assistant

- Served as Teaching Assistant in Fluid Mechanics, Thermodynamics I/II, and Process design I/II.

Didas International Group/Iran (2009–2010)
Intern

- Prepared technical reports and presented key insights to internal and external stakeholders.

Honors and Awards

- Recipient of Teaching Assistant of the Year Award at IIT (2016 & 2017).
- Recipient of Graduate Assistantship, Illinois Institute of Technology, Fall 2012.
- Recipient of Graduate Assistantship, University of Texas A&M-Kingsville, Fall 2010.
- Recipient of Full Scholarship, Sharif University of Technology, Fall 2005.

Affiliations

- Member of American Institute of Chemical Engineers (2014-present).
- Liaison of ChBE Ethics Committee (NSF-funded project, 2018)
- President of Graduate Student Association at ChBE department, IIT (2016-2017).
- Member of Society of Petroleum Engineers (2010-2012).
- Member of executive student committee of the 2nd and 3rd HSE conference at SUT (2009-2010).
- Representative of petroleum engineering students at Sharif University of Technology (2005-2010).